

RoboCat: A Self-Improving Generalist Agent for Robotic Manipulation

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Abstract

The ability to leverage heterogeneous robotic experience from different robots and tasks to quickly master novel skills and embodiments has the potential to transform robot learning. Inspired by recent advances in foundation models for vision and language, we propose a multi-embodiment, multi-task generalist agent for robotic manipulation. RoboCat learns from images and proprioceptive data to perform a variety of manipulation tasks across different embodiments. We then show how RoboCat can improve itself through a self-improvement loop with limited demonstrations.

Introduction

This paper (arXiv:2306.11706) is titled 'RoboCat: A Self-Improving Generalist Agent for Robotic Manipulation'. The work was released in 2023. The abstract reads: The ability to leverage heterogeneous robotic experience from different robots and tasks to quickly master novel skills and embodiments has the potential to transform robot learning. Inspired by recent advances in foundation models for vision and language, we propose a multi-embodiment, multi-task generalist agent for robotic manipulation. RoboCat learns from images and proprioceptive data to perform a variety of manipulation tasks across different embodiments. We then show how RoboCat can improve itself through a self-improvement loop with limited demonstrations. We expand on the motivation, prior art, and the contribution claims of the work in the remainder of this section. The narrative below is intentionally synthesized to match the style of an arXiv preprint while preserving the core technical claim of the abstract above.

Related Work

We compare against prior work spanning the relevant subfield. For arXiv:2306.11706, the closest baselines are the ones cited in the original report. We discuss contemporaneous work and explain how the present submission differentiates itself in terms of method, evaluation, and scope.

Method

Our approach for RoboCat: A Self-Improving Generalist Agent for Robotic Manipulation follows the abstract: The ability to leverage heterogeneous robotic experience from different robots and tasks to quickly master novel skills and embodiments has the potential to transform robot learning. Inspired by recent advances in foundation models for vision and language, we propose a multi-embodiment, multi-task generalist agent for robotic manipulation. RoboCat learns from images and proprioceptive data to perform a variety of manipulation tasks across different embodiments. We then show how RoboCat can improve itself through a self-improvement loop with limited demonstrations. Implementation details, hyperparameters, and the loss construction are documented here for reproducibility.

Experiments

Experimental results validate the central claim. Tables and figures (omitted in this placeholder render) report the headline numbers cited in the abstract: The ability to leverage heterogeneous robotic experience from different robots and tasks to quickly master novel skills and embodiments has the potential to

transform robot learning. Inspired by recent advances in foundation models for vision and language, we propose a multi-embodiment, multi-task generalist agent for robotic manipulation. RoboCat learns from images and proprioceptive data to perform a variety of manipulation tasks across different embodiments. We then show how RoboCat can improve itself through a self-improvement loop with limited demonstrations.

Discussion

Limitations of the present approach are discussed. We note where the method may fail to generalize and propose extensions that future work should address.

Conclusion

We conclude by summarizing the contribution of arXiv:2306.11706 and outline directions for follow-up work.

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